

Boston University Electrical & Computer Engineering

Customer Installation Report



Team 25

1.0 Customer Installation Details:

For the preliminary installation, all team members were present on Friday April 25, where they met with Professor Lagoy and an associate of his named Noah Bagazinski, a graduate student at MIT. Noah provided a sailboat for the team to do a preliminary installation and testing with a more formal and final installation planned to happen on April 29 with Kay VanValkenburgh on a larger vessel. The ship that was used with Noah was much smaller with no railings or lifelines for our system to be properly mounted.

2.0 Requirements:

Requirement	Type	Modification (if applicable)	Completion (if applicable) (Y/N)
The sensor system shall report proximity to nearby objects and boats via an audio and visual alert.	Modified	Only visual alerts	Y
The sensor system shall report speed over ground, speed through water, wind speed/direction.	Original	N/A	Y
The system shall include real-time collision avoidance alerts, and as a bonus, follow right-of-way (COLREGs) rules.	Original	N/A	Y
The sensor system shall be easily installed/removed from all types of boats without causing damage.	Original	N/A	Y
The sensor system shall have reliable BLE communication with an iPhone.	Original	N/A	Y
The system shall contain multiple sensor types: a. Distance sensors (with Object Detection)	Original	N/A	Y

b. GPS, compass, speed-over-ground c. Wind speed (Anemometer)			
The system shall be powered via a rechargeable battery, as most small sailboats don't have accessible power.	Original	N/A	Y
The status shall be reported in a visually pleasing iOS app.	Original	N/A	Y
The app shall report to the user recommended changes in course based on sensor readings.	Skipped/Eliminated	N/A	N
The system shall be designed to be dust/waterproof rated to IP68 and able to withstand harsh marine environments.	Modified	IP66 rating	Y

3.0 Assessment:

In our testing and installation the team showcased a live demo of the Triton being used on a sailboat in real testing conditions. There was a last-minute complication with the StereoPi side cameras that did not allow them to be deployed on the ship, however for the second installation period with Kay on April 29th they will be fixed and properly deployed on the sailboat.

Overall, the product worked as intended. The housing was secure and sturdy, with a method to quickly insert the sensors in a manner that does not alter or damage the ship and ensure the security of the devices. Additionally, the housing was printed using a special filament with waterproof properties and coated in a sealant. The iOS application was able to seamlessly connect to the device and receive the data from the sensors and OAK-D camera in a clean and readable manner. The UI had various pages for different information based on what the user was most interested in seeing at the time. In addition to this, the rendering was able to accurately depict the distances detected by the OAK-D camera as well as the objects the camera could recognize (boats and other people). Finally, the OAK-D was able to successfully record the distances of different objects and produce accurate depth readings, although sometimes the waterline would be detected. This was fixed when a different script was run that had been pushed to the project GitHub after the most recent pull before the testing ensued.

The second session of testing and installation aims to be a final test for the product. The goals for this followup session will be to fully ensure the functionality of the mounts for the housing on the front and side of the ship. Additionally, the OAK-D's waterline recognition and avoidance will be fully implemented and completed, as well as the implementation for the StereoPi cameras on the side. The product has exhibited the necessary qualities and features as desired by the client, and this final session aims to showcase the full extent of the Triton.

4.0 Customer Letter:

After the testing had concluded, Professor Lagoy emailed the team with his thoughts on the testing and hopes for our second round of final implementation:

“I wanted to give you some initial feedback for your automatic docking assist and sailboat detection system.

Specifically, the app layout and docking screen is exactly as I imagined. Additionally, the Oak D stereo cam and RGB cam inputs were effectively combined to provide both distance alerts and sailboat detection. The wind speed sensors worked perfectly on our demo. I'm looking forward to seeing the side stereo camera sensors!

Overall, I'm very impressed with how you integrated all the subsystems into a thoughtfully designed enclosure with focus on safety, ease of use and aesthetic. Great work team!”